

$$= \sum_{k=1}^m (b_{ik} \cdot a_{kj} + c_{ik} \cdot a_{kj}) = \sum_{k=1}^m b_{ik} \cdot a_{kj} + \sum_{k=1}^m c_{ik} \cdot a_{kj}$$

The (i,j) th entry of BA is: ~~$b_{i1} \cdot a_{1j} + b_{i2} \cdot a_{2j} + \dots + b_{im} \cdot a_{mj}$~~
 $b_{i1} \cdot a_{1j} = \boxed{\sum_{k=1}^m b_{ik} \cdot a_{kj}}$

The (i,j) th entry of CA is: $c_{i1} \cdot a_{1j} + c_{i2} \cdot a_{2j} + \dots + c_{im} \cdot a_{mj}$
 $= \boxed{\sum_{k=1}^m c_{ik} \cdot a_{kj}}$

$\therefore$ The (i,j) th entry of $(B+C)A$ is the sum of the (i,j) th entry of BA and CA .

$$\therefore (B+C)A = BA + CA \quad \square$$

20) Prove Theorem 2(d).

Solution: $r(AB) = (rA)B = A(rB)$ for any scalar r .

The (i,j) th entry of AB is $a_{i1}b_{1j} + a_{i2}b_{2j} + \dots + a_{in}b_{nj}$
 (By Row-Column Rule)
 $= \sum_{k=1}^n (a_{ik} \cdot b_{kj})$

$\therefore$ The (i,j) th entry of $r(AB)$ is $r \sum_{k=1}^n (a_{ik} \cdot b_{kj}) = \sum_{k=1}^n r(a_{ik} b_{kj})$
 (By Row-Column Rule)
 $= \sum_{k=1}^n (r a_{ik}) \cdot b_{kj}$

The (i,j) th entry of $(rA)B$ is $r a_{i1} \cdot b_{1j} + r a_{i2} \cdot b_{2j} + \dots + r a_{in} \cdot b_{nj}$
 $= \sum_{k=1}^n (r a_{ik}) \cdot b_{kj}$
 $\therefore r(AB) = (rA)B \quad \square$

$\sum_{k=1}^n r(a_{ik} b_{kj}) = \sum_{k=1}^n (r a_{ik}) b_{kj} = \sum_{k=1}^n (a_{ik} r) b_{kj} = \sum_{k=1}^n a_{ik} (r b_{kj})$
 (By Row-Column Rule)

The (i,j) th entry of $A(rB)$ is $a_{i1} \cdot (r b_{1j}) + a_{i2} \cdot (r b_{2j}) + \dots + a_{in} \cdot (r b_{nj})$
 $= \sum_{k=1}^n a_{ik} \cdot (r b_{kj})$
 $\therefore r(AB) = (rA)B = A(rB) \quad \square$

21) Show that $\text{Im } A = A$ when A is an $m \times n$ matrix. You can assume $\text{Im } \lambda = \lambda$ for all λ in $\mathbb{R}^m$.